Binita Bhusal

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SUMMARY

Final-year BIT student at Tribhuvan University with hands-on experience building production-grade Django REST Framework backends. Built CivicAid, a full civic complaint management platform with JWT authentication, an ML pipeline for automated classification, and async notifications deployed on Render with a React frontend on Vercel. Also built a multi-role Job Portal backend with a full application lifecycle, Celery + Redis async notifications, and Swagger API documentation. Seeking a backend development role to grow toward AI-integrated systems.

TECHNICAL SKILLS

Backend: Python, Django, Django REST Framework, JWT Authentication, REST API Design, Celery, Redis

ML / Data: scikit-learn, NLTK, TF-IDF, Naive Bayes, Cosine Similarity, NumPy, Pandas (basics)

Databases: PostgreSQL, database indexing, query optimisation (select_related, prefetch_related)

Frontend: React.js, Tailwind CSS, React Router DOM

Languages: Python, JavaScript, C, C++, C# (familiar)

Soft Skills: Problem Solving, Team Collaboration, Quick Learner

Tools: Git, GitHub, Postman, Docker (basics), drf-spectacular / Swagger, Cloudinary, Brevo, Neon.tech, Render, Vercel, Vite, SQL

PROJECTS

CivicAid — *Django, DRF, scikit-learn, Celery, PostgreSQL, Cloudinary, Brevo*

- Built the complete backend independently: 6 models, JWT authentication, role-based permissions (Citizen / Admin), and a REST API consumed by a React frontend deployed on Vercel
- Integrated an ML pipeline using TF-IDF + Naive Bayes for automatic complaint category (~68% accuracy) and priority (~67% accuracy) classification
- Implemented cosine similarity-based duplicate complaint detection, with results persisted to the Complaint model via perform_create()
- Added in-app and email notifications via Brevo, OTP-based email verification, heatmap and admin trend endpoints, and database indexes on status, category, and created_at for query performance
- Deployed on Render (backend) and Vercel (frontend); media stored on Cloudinary for persistence across dyno restarts

GitHub: github.com/bbinita/civicaid

Job Portal API — *Django, DRF, PostgreSQL, Celery, Redis, Docker, drf-spectacular*

- Built a multi-role backend (Company, Candidate, Admin) with custom permission classes and JWT authentication
- Designed a full application status lifecycle (SUBMITTED → UNDER_REVIEW → SHORTLISTED / REJECTED → SELECTED) with model-layer transition validation to prevent illegal state changes
- Optimised queries using select_related and prefetch_related to eliminate N+1 problems; added db_index on status and applied_at fields
- Integrated Celery + Redis for async email notifications on status changes; ran Redis in Docker during development
- Generated interactive Swagger API documentation with drf-spectacular; wrote pytest-django tests covering status transition logic

GitHub: github.com/bbinita/job-portal

Resume Screener — *Django, pdfplumber, NLTK, scikit-learn, TF-IDF*

- Extracts text from uploaded PDFs and screens resumes using TF-IDF and cosine similarity against a job description
- Integrated NLTK for text preprocessing and scikit-learn for similarity scoring

GitHub: github.com/bbinita/resume-screener

Chatterly — Real-Time Chat Rooms — *Django, WebSockets, Django Channels*

- Built a real-time chat application with dynamic room creation and multi-user live messaging without page reloads using Django Channels and WebSockets

GitHub: github.com/bbinita/Chatterly-Real-Time-Chat-Rooms-in-Django

EDUCATION

Bachelor of Information Technology (BIT) | Tribhuvan University, Bhairahawa Multiple Campus

2021 – 2026 • Expected graduation: 2026 • 7th of 8 semesters completed

High School (+2 Science) | Kathmandu Model College

2020 – 2021

CERTIFICATIONS

Responsive Web Design — freeCodeCamp

JavaScript Algorithms and Data Structures — freeCodeCamp